

Final Project

This is the second of two projects that make up the assessed part of the course. This project is worth 85% of the total course mark for SPA7037U/P and 50% of the total course mark for SPC707P. The deadline for submission is April 15th 2026 at 23:59. All three files must be submitted to QMPlus to receive marks.

What you should submit

- A written report in pdf format (page limit 5 pages, minimum font size 11pt) on the project which should also contain the specific plots requested. This should be in the form of a complete scientific report that you might find in a journal. Anything that you want to be graded should be included in the project report document.
- Your python + tensorflow code in a jupyter notebook that should run on the Jupyter-Hub profile for the course
- Your python + tensorflow code exported to a separate pdf

Regression

You have a choice of dataset in this project, however try not to use any we have used in class or any that your colleagues are using as only original work can get marks. The design and training of your network must be your own original work and for the removal of doubt this excludes the use of transfer learning. You must use real world datasets, for this project (not generated ones) although data augmentation is okay. Kaggle datasets are banned and will result in your grade being capped at the pass mark. Design and train a neural network to perform the following:

- One non-linear regression task of your choice

To perform this task use one of the following three methods we have met in class:

- MLP
- CNN
- RNN

Make sure that whatever network you choose is appropriate for the task you are trying to perform.

- Explain your choice of problem and dataset. State where you obtained your dataset (reference). Show a small but representative selection of your input data before and after, if appropriate, pre-processing and explain why the preprocessing was necessary (if it was, otherwise explain why it wasn't).
- Explain your choice of network to solve this problem. Include a diagram of your network design and explain the choice of network architecture and activation functions in each layer.
- Now train the network to perform the task.
- Include a plot of your training and test loss as a function of number of training epochs and explain how you check that your network is not over-fitting.
- Make a plot of your true test labels against your model's predictions on the test set. Make the same plot but showing the residual differences.
- Make a different plot that shows the performance of your network on your test data.
- Has the network performed well? How can you tell? Are there any specific aspects of the performance that could be improved?